EARTH AND PLANETARY SCIENCE LETTERS 18 (1973) 37–44. NORTH-HOLLAND PUBLISHING COMPANY

PALEOMAGNETISM OF LATE CRETACEOUS POÇOS DE CALDAS ALKALINE COMPLEX, SOUTHERN BRAZIL*

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Received 19 September 1972
Revised version received 1 December 1972

The paleomagnetism of the Late Cretaceous Poços de Caldas alkaline complex (46.6°W, 21.9°S) was investigated through 42 oriented cores from seven sites. Six sites, reversed relative to the present magnetic field of the Earth, yield a pole at 127°W, 82°S ($dp = 8°$, $dm = 13°$). This pole is located close to other Late Cretaceous poles for South America obtained by Creer [1] from untreated paleomagnetic samples. The results are significantly different from those for the nearby Early Cretaceous Serra Geral basalt but close to the Triassic pole for South America. The polar wandering path for South America for the Mesozoic seems to be more complicated than anticipated. The available paleomagnetic information may not yet be precise enough to determine the time of opening of the Atlantic.

1. Introduction

Paleomagnetism of Jurassic and Cretaceous rocks of South America and Africa probably records the opening of the South Atlantic. Unless such opening was through a rotation about a pole close to the axis of rotation of the Earth, the time of this tectonic phenomenon should be recorded by a fork in the superimposed polar wandering curves of South America and Africa. This report presents new paleomagnetic information on the Late Cretaceous of South America and evaluates it in relation to other available data. Oriented cores were taken with the assistance of I. Pacca of the Institute of Physics of the University of São Paulo. Paleomagnetic laboratory measurements were done at Lamont-Doherty Geological Observatory of Columbia University.

2. Geological setting

East of the Paraná basin of southeast Brazil are numerous late Mesozoic alkaline extrusive-intrusive

* Lamont-Doherty Geological Observatory Contribution No. 1866

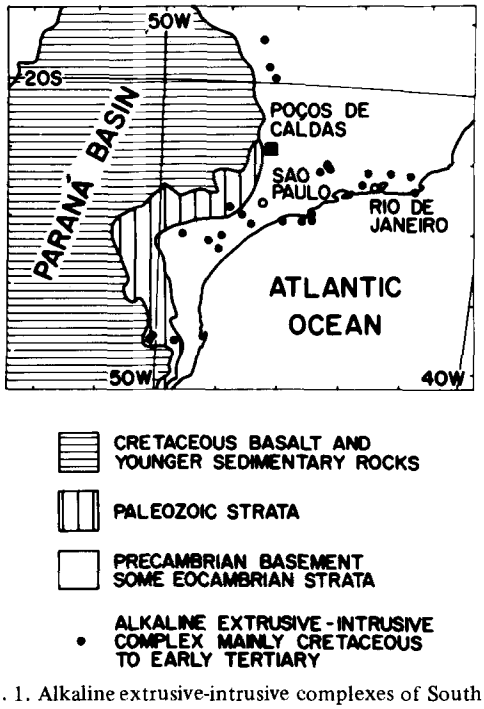


Fig. 1. Alkaline extrusive-intrusive complexes of South Brazil, after Amaral et al. [2]. Small square outlines location of Poços de Caldas complex of Late Cretaceous age.

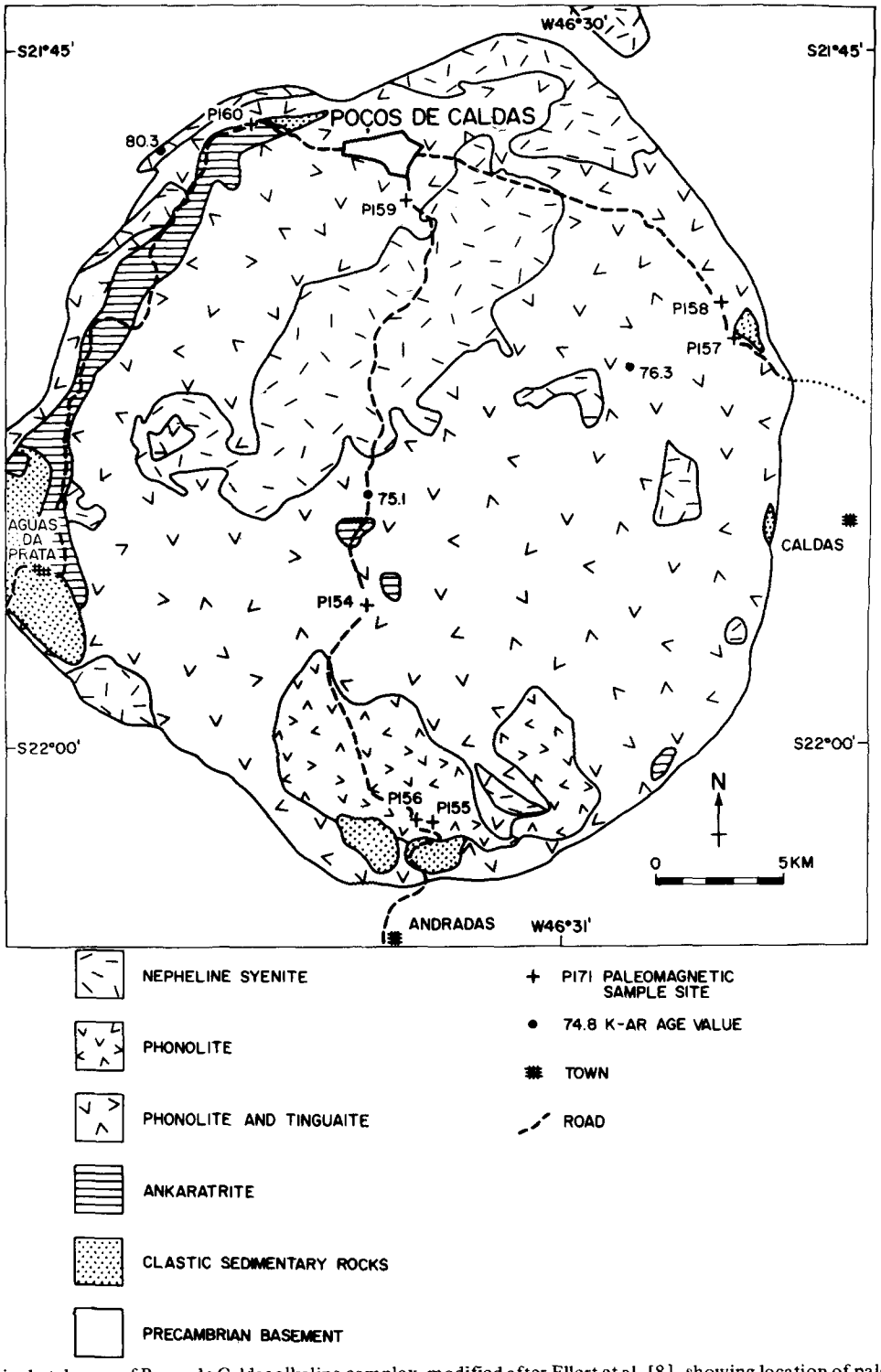


Fig. 2. Geologic sketch map of Poços de Caldas alkaline complex, modified after Ellert et al. [8], showing location of paleomagnetic sample sites.

complexes (fig. 1). These fall into two age groups: 133 to 122 my (Early Cretaceous) and 82 to 51 my (Late Cretaceous to Early Tertiary) ([2]; see also [3, 4]). The older group, lying southwest of the yonger group, overlaps in time the extrusion of the widespread Serra Geral plateau basalt, the paleomagnetism of which has been studied by Creer [5]. The largest complex of the younger group is the Poços de Caldas complex underlying an area of 800 km^2. Topographically, this complex underlies a rimmed circular basin which rises above the surrounding Late Precambrian terrain. Contributions to the petrology and structure of this feature have been made by Ellert [6], Bjornberg [7] and Ellert et al. [8]. The complex consists of several lithologically distinctive units (fig. 2).

The most abundant rock-types are phonolite (the fine-grained equivalent of nepheline syenite and its hypabyssal equivalent, tinguaite. The phonolite and tinguaite post-date ankaratritic extrusives (similar to nepheline basalt) in the west and are intruded by nepheline syenite in the central and northern parts of the structure. Blocks of unfossiliferous clastic sedimentary rocks lithologically similar to the late Mesozoic Botucatú sandstone have been engulfed by the alkaline igneous rocks. These inclusions are preserved near the periphery of the complex. Because of deep weathering, the internal structural details of the complex are not well-known. The phonolite and tinguaite form massive outcrops mostly lacking well-defined layering. Faults have not been mapped in detail. Concentric faults near the periphery of the complex are suspected, and numerous mineral springs are known in the border zone. The presence of syenite stocks near the center and northeast regions and of subsided blocks of clastic rocks near the periphery has been noted. This distribution of units suggests uplift of the central and northeast parts of the complex and relative subsidence of the outer regions. No Cretaceous or younger orogenic folding of possible consequence to paleomagnetic studies is evident. After emplacement of the complex erosion stripped away the surrounding Botucatú sandstone, leaving the complex isolated in the Precambrian basement.

The age of the complex is known from the studies of Amaral et al. [2] and from unpublished data of Bushee, who carried out K-Ar age analyses at the University of São Paulo and at the University of California at Berkeley. Amaral et al. [2] reported ages near 75 my for the central area of tinguaite and phonolite. This Late Cretaceous age, approximately Santonian to Campanian according to the time-scale of the Geological Society of London [9], is thought to be representative of six of the sites of the present study (P-154 to P-159). One site (P-160) lies in the ankaratritic unit, which may be slightly older.

3. Sampling

Five to seven oriented cores were taken at each of seven sites by means of portable equipment which cuts cores 2.54 cm in diameter. Most sites are located in road cuts in the outer part of the complex, where topographic relief is greater and exposures are consequently better. One site (P-159) is located in a stream-bed at the south limits of the city of Poços de Caldas.

All cores were oriented by magnetic compass. Sun-compass checks were possible on 11 cores; in most cases magnetic azimuths of the cores agree with sun-compass azimuths within ± 3°. Where the disagreement was more than 3° the sun-compass orientations were used. No bedding corrections were possible. The fairly small scatter in the data suggests that such corrections would be small, if necessary. Therefore, the present horizontal is taken to be the same as the paleo-horizontal.

After being returned from the field, the samples were cut into cylinders 2.54 cm in length and measured on a commercially available 100 cps spinner magnetometer. Susceptibility measurements were carried out on a commercially available a.c. bridge.

4. Natural remanent magnetization observations

The intensity of magnetization for sites P-154 to P-159 ranged from 1.4×10^{-5} to 9.3×10^{-5} emu/cm. The average remanent intensity of site P-160 was much higher than for the other sites, being 6.6×10^{-3} emu/cm^3. The bulk susceptibilities were one order of magnitude less than the intensity of remanent magnetization and for sites P-154 to P-159 the average bulk susceptibilities ranged from 1.6×10^{-6} to 7.2×10^{-6} emu/cm^3. The susceptibility for site P-160 was much higher than for the other

Table 1
Mean NRM and a.f. demagnetized directions

Site	Lat. (°S)	Long. (°W)	N	n	NRM D_{NRM} (°)	NRM I_{NRM} (°)	NRM k	NRM α_{95} (°)	a.f. (Oe)	after a.f. demagnetization D_{af} (°)	after a.f. demagnetization I_{af} (°)	after a.f. demagnetization k	after a.f. demagnetization α_{95} (°)	MDF (Oe)
P-154	21.9	46.6	6	11	14.1	+49.6	8.0	17.1	1000	154.5	+64.3	17.0	12.0	125
P-155	22.0	46.5	6	9	189.6	+35.5	131.2	4.5	150	187.6	+28.7	962.8	1.9	350
P-156	22.0	46.6	6	11	196.8	+35.5	13.0	13.1	150	191.4	+35.2	21.0	10.7	400
P-157	21.9	46.4	6	9	251.0	+13.8	89.3	5.4	300	213.7	+42.5	2.4	37.7	30
P-158	21.8	46.4	5	5	19.9	−65.6	97.4	7.7	200	173.6	+39.1	43.8	14.0	75
P-159	21.8	46.6	6	10	226.5	+49.1	3.2	31.8	200	178.1	+45.3	84.0	6.0	180
P-160	21.8	46.6	6	7	7.7	−37.2	27.2	11.7	300	17.2	−48.7	13.8	13.4	225

N is the number of cores drilled in the field; n is the number of samples cut from the cores used in the analysis. Statistics are after Fisher [20]. MDF is the median destructive field, which is the alternating field necessary to reduce the remanent intensity by 50%.

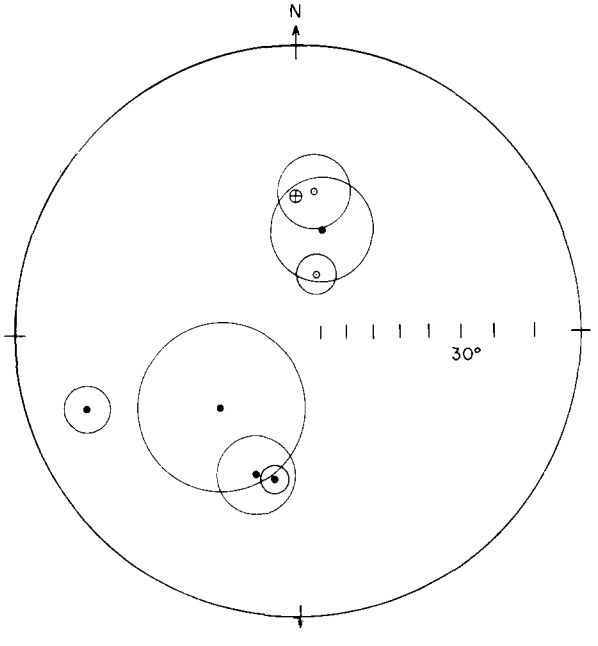


Fig. 3. NRM directions of magnetization for Poços de Caldas complex site-means (table 1). Closed circles are lower-hemisphere projections. Equal angle projection with local horizontal as primitive. Circles of 95% confidence are drawn around each site-mean direction. The cross enclosed in the circle is the direction of the axial dipole field.

sites, having an average value of 3.2×10^{-4} emu/cm^3.

The directions of NRM are given in table 1 and are shown in fig. 3. It can be seen that the within-site precision is quite good for the various sites, with k values as high as 131 for site P-155. The site-mean directions, however, are highly dispersed, with four inclined below the horizontal in the southwest quadrant. The other site-mean directions have northerly declinations; two have negative (upward) inclinations and one has a positive inclination. The site-mean direction for site P-160 is not significantly different from the present direction of the earth's field at the sampling locality.

5. Alternating-field demagnetization results

Two samples from each site were subjected to step-wise a.f. demagnetization up to or above 200 Oe peak field. The a.f. demagnetization apparatus is of the type described by McElhinny [10].

One a.f. demagnetization curve from each site is shown in fig. 4. It can be seen by the shapes of a.f. demagnetization curves that a wide spectrum of remanent coercivities is represented. This is shown by the wide range of values for the median destructive field (table 1), from 30 to over 400 Oe. An interesting a.f. demagnetization curve is that of a specimen

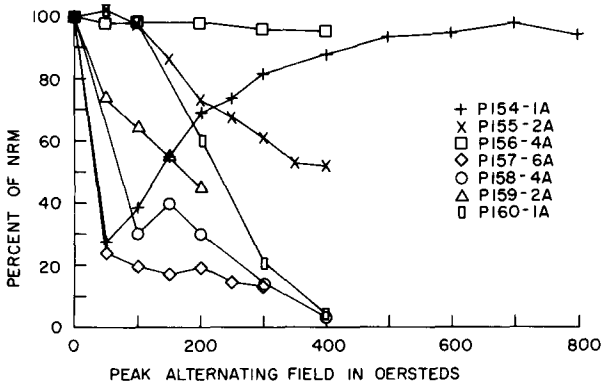


Fig. 4. Alternating-field partial demagnetization curves of representative samples from paleomagnetic sites in the Poços de Caldas complex. Percent intensity of magnetization of sample relative to its NRM is plotted as ordinate. A.c. demagnetizing field strength in oersteds is plotted as abscissa.

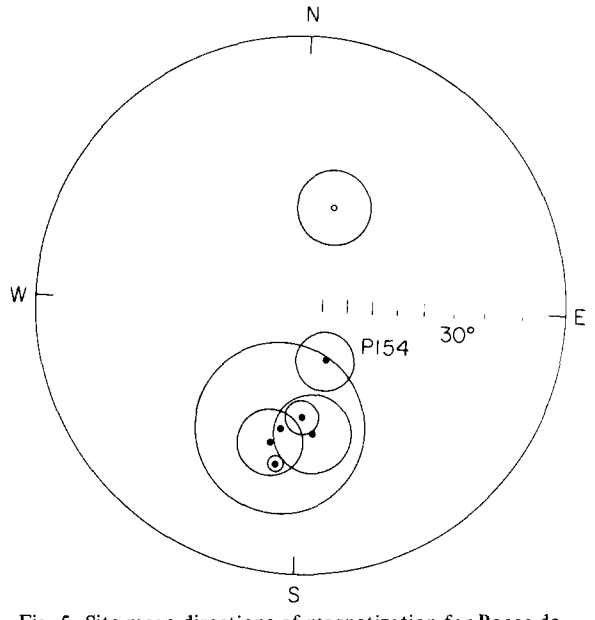


Fig. 5. Site-mean directions of magnetization for Poços de Caldas complex after a.c. demagnetization (table 1). Closed circles are lower-hemisphere projections. Equal angle projection with local horizontal as primitive. Circles of 95% confidence are drawn around each site-mean direction.

from site P-154, which shows an initial sharp decrease in lower fields and a steady rise in intensity of magnetization with higher fields. The a.f. demagnetization curve shown is interesting, in that fields above 400 Oe, almost the entire intensity of NRM is recovered. Other specimens from this site do not have such a sharp initial drop and recover only about one-third of the initial NRM intensity. This behavior is clearly the result of secondary components of varying stability directed in the direction of the present field. In these specimens the resistance of these components to a.f. demagnetization indicates the possibility that the high-coercivity secondary component is due to the growth of hematite in the present field. The stability is too high to be due to simple VRM.

The mean directions of magnetization for each site after a.f. demagnetization in fields of peak value from 150 to 1000 Oe are given in table 1 and are plotted in fig. 5. Six of the sites yield mean directions towards the south with positive inclinations. These sites are reversed with respect to the present field direction and most are well-grouped. The least stable site with the largest circle of confidence is site P-157. This site is actually better grouped before a.f. demagnetization than after (table 1). All specimens from this site were demagnetized in steps up to 400 Oe. The preferred values, obtained in a washing field of 300 Oe, are marginally better than at other field intensities. Sites P-155, P-156 and P-159 were initially directed downwards towards the south. After a.f. demagnetization, the scatter of vector orientations decreases, and the mean directions move closer together. Site P-158 was initially magnetized towards the north and up, but upon a.f. demagnetization, the site-mean direction moves through 180° and joins the reversed group. Site P-154 moves towards the reversed group of site-mean directions and was still moving when a.f. demagnetization was terminated at a field strength of 1000 Oe. The moste stable component from this site is clearly reversely magnetized, but because the directions of magnetization did not reach a stable position, the results have been excluded in the calculation of the pole position for the complex. Although site P-160 is normally magnetized and remains so on demagnetization, this site lies within the ankaratritic unit which, as was noted above, may be somewhat older than the phonolite and tinguaite units. The direction of the mean vector of site P-160 is reversed for the calculation of the mean pole position, which is found at 127°W, 81°S (dp = 8°, dm = 13°) (table 2).

Table 2
Paleomagnetic pole data for Late Cretaceous Poços de Caldas alkaline complex, Brazil

						Pole position			
N	R	D	I	k	α_{95}	Lat.	Long.	dp	dm
6	5.87	190.0°	40.6°	39.2	10.8°	80.7°S	127.2°W	7.9°	13.1°

N = number of sites; R = length of mean vector; D = declination of mean vector; I = inclination of mean vector; k = kappa value of Fisher distribution; α_{95} = half-angle of cone of 95% confidence around mean vector; dp = semi-axis of 95% oval of confidence along a great circle joining the site with the paleomagnetic pole; dm = semi-axis of 95% oval of confidence in the direction perpendicular to the preceding.

6. Interpretation of the results

With the exception of work on the Serra Geral basalt of Brazil and of the La Teta lava of Colombia, no other a.f. or thermally demagnetized results on oriented samples of Cretaceous rocks of South America are known. Our results are significantly different from those for the Early Cretaceous Serre Geral basalt (fig. 6a) studied by Creer [5]. The La Teta lava appears to have been rotated tectonically [11], so that a direct comparison of poles is of little value in evaluating the position of the Cretaceous pole for South America.

The results from the Poços de Caldas are, however, not significantly different from results obtained from the Tertiary Boqueron sediments by Creer [12].

The Late Cretaceous confidence oval reported here overlaps that of the South American Triassic, estimated by Valencio and Vilas [13] to be near 129°W, 79°S (α_{95} = 16°). The distribution of the Triassic, Jurassic [14, 15], Early Cretaceous and Late Cretaceous paleomagnetic poles (fig. 7) suggests a more or less oscillatory motion of the paleomagnetic pole for South America from Triassic to Late Cretaceous time. This makes the use of paleomagnetism for age identification of South American Mesozoic rocks difficult and indicates a more complex late Mesozoic polar wandering path than had been anticipated.

It is impossible at this time to compare our results

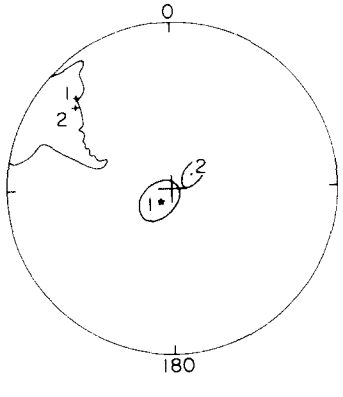


Fig. 6. Comparative Cretaceous paleomagnetic pole positions for South America. Equal area projection. Sampling sites are shown by crosses. The mean pole positions are plotted along with ovals of 95% confidence. The numbers refer to those in table 3.

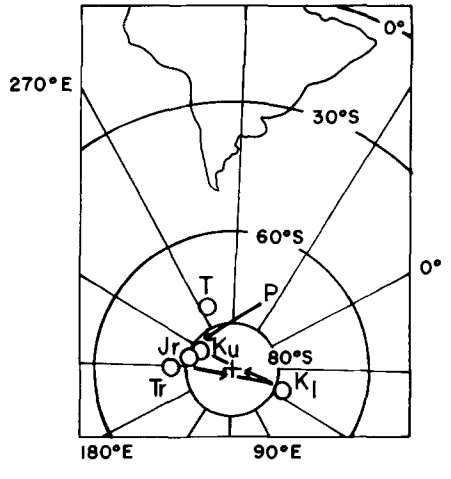


Fig. 7. The apparent polar wander path for South America during the Mesozoic to Late Cretaceous time plotted on a polar equal angle projection.

Table 3
Comparative Cretaceous pole positions for South America and Africa

		Site location		Pole location				Rotated Site location		Rotated pole location		
	Pole	Lat.	Long.	Lat.	Long.	dp	dm	Lat.	Long.	Lat.	Long.	Ref.
Poços de Caldas	1	21.9°S	46.6°W	80.7°S	127.2°W	7.9°	13.1°					this paper
Sierra Geral fm	2	25°S*	50°W*	78.0°S	54.0°E	4°	6.7°					[5]
Lupata series	3	16.8°S	34.2°E	61.8°S	79.5°E	3.2°	4.5°	41.0°S	26.6°W	76.6°S	138.3°W	[21]
Malanje Intrusives	4	16.0°S	35.5°E	60.0°S	82.0°E	9.1°	13.0°	41.1°S	24.7°W	78.4°S	144.1°W	[22]

* Approximate location for many scattered sites.

with African results of the same age, since results from Africa for the Upper Cretaceous are not available. The youngest Cretaceous-yielding results from Africa, the Lupata volcanics reported by Gough and Opdyke [21], are 35 my older than the Poços de Caldas complex. Evidence from the South Atlantic obtained from anomaly studies [16, 17] and from the Deep Sea Drilling Project [18] indicates that the Atlantic had opened to about one-half of its present width by the time of the emplacement of the Poços de Caldas complex (75 my). Allard and Hurst [19] have placed the time of opening of the South Atlantic as mid-Cretaceous, based on the similarities of the stratigraphic sections of Gabon and Brazil. Paleomagnetic information from South America [15] and Africa from the Jurassic indicates that the South Atlantic was closed at that time. Paleomagnetic results from the Lower Cretaceous of the Serra Geral of Brazil and the mid-Cretaceous of Africa do not overlap on closure of the South Atlantic, perhaps indicating that the South Atlantic was open by this time. Further work is necessary in both Africa and South America to satisfactorily solve this problem.

7. Conclusions

A Late Cretaceous virtual geomagnetic pole for the Poços de Caldas complex of Brazil (46.6°W, 21.9°S) is located at 127°W, 81°S (dp = 8°, dm = 13°). The Late Cretaceous pole for South America derived from the Poços de Caldas complex is significantly different from Creer's [5] Early Cretaceous pole, indicating a significant shift, possibly through 15°, for the South American pole in mid-Cretaceous time. The Late Mesozoic polar wandering curve for South America is more complicated than anticipated. The time of the separation of South America and Africa, as judged from branching of their respective superimposed polar wandering curves, requires more data, especially from the Cretaceous of both continents.

Acknowledgements

We are grateful to J.M. Coutinho and the Research Foundation of the State of São Paulo for making possible the visit of one of us (WDM) to Brazil, and for the support of the National Science Foundation, grant GA-1548. Stimulating conversations with U. Cordani and G. Amaral of Brazil and D. Valencio of Argentina have been helpful in the collection of samples and analysis of data. To I. Pacca and the Institute of Physics of the University of São Paulo we are indebted for invaluable assistance in the field. G. MacDonald assisted generously in the field and with the computer programming. We also would

like to thank M.W. McElhinny and K.M. Creer for reading the manuscript and making constructive comments.

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